TED (15) – 4064 (REVISION – 2015)

FOURTH SEMESTER DIPLOMA EXAMINATION IN TEXTILE TECHNOLOGY

YARN FORMATION - II

(Maximum Marks : 100)

Time : 3 Hours

PART – A

(Maximum marks: 10)

I Answer the following questions in one or two sentences. Each question carries 2 marks:

Marks

1. State the functions of auto levelers in Draw frame process.
2. List the various top roller weighting systems in Draw frame.
3. State the functions of building motion in a fly frame.
4. List the objectives of Ring frame.
5. Find the hank of lap of a sliver lap if it weighs 60 grams per meter (5 x 2 =10)

PART – B

(Maximum marks : 30)

II Answer any five questions. Each question carries 6 marks:

1 Discuss the importance of stop motion in Draw Frame.

2 Discuss the functional parts of comber .

3 List any three roving defects - their causes and remedies.

4 Describe the twisting operation in fly frame.

5 Describe the method of producing ‘Z’ twist and ‘S’ twist yarn.

6. State the features of pneumatic weighting systems in Ring frame.

7 Find the actual production of sliver lap machine per shift of 8 hours in kgs :
Speed : 40 mpm, Hank lap : 0.014, Efficiency : 85%

(5 x 6 = 30)

PART – C

(Maximum marks : 60)

(Answer one full question from each unit . Each question carries 15 marks)

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UNIT – I

III. (a). State the terms floating fibers and drafting waves in draw frame. 5

(b) Explain the working of Ribbon lap machine with sketch . 10

OR

IV (a) State the need of top roller cleaning devices in draw frame 5

(b) Explain the passage of material through a modern comber with neat sketch . 10

UNIT – II

V (a) List the top roller weighting systems employed in fly frame .and describe any one. 5
(b) Explain the working of Fly frame with sketch . 10

OR

VI (a) List the advantages of apron drafting system in fly frame . 5
(b) Discuss the working of differential motion in fly frame with sketch. 10

UNIT – III

VII (a) List any five causes of end breaks in Ring Frame. 5
. (b) Sketch and explain the working of building motion for weft wind. 10

OR

VIII (a) Explain the salient features of SKF-PK225 drafting system in Ring frame. 5
(b) Explain the methods of twist insertion and winding in ring frame with sketch. 10

UNIT – IV

IX (a) Draw the draft gearing of Drawing frame and Calculate Draft constant
Diameter of front and back roller – 1 inch , Front roller wheel – 28 T ,
Crown wheel – 100 T , Change pinion – 32T , Back roller wheel - 80 T. 7

(b) Calculate the production of a speed frame in Kgs/shift of 8 hours from the given data
Spindle speed – 1000rpm , Roving hank – 2.0, TM – 1.1 , Efficiency -85 % , .
Number of spindles -120. 8

OR

X (a) Calculate the production of a high speed Draw frame in Kg per shift of 8 hours with.
the following data
Front roller delivery– 420 m/min, Number of delivery – 2, Number of ends up - 8
Efficiency – 86 % , Draft – 8.2 , Card sliver hank – 0.13. 7

(b) Calculate the production of a comber in Kg /shift of 8 hours with following data
Nips/min -200 Feed roller diameter -30mm Feed ratchet wheel -14T
No. of tooth pick/nip -1 Noil -15% Lap fed -55grams/meter Efficiency -85%
8